Sheet 10

Pointer noise, finite disturbance and Zeno

1. For the centred chirped meter $\varphi_b(y) = (2\pi\sigma_M^2)^{-1/4} e^{-(1-ib)y^2/(4\sigma_M^2)}$, with $b \in \mathbb{R}$ and $\sigma_M > 0$, compute its position and momentum spreads, then ϵ_Q, η_P for the lecture's impulsive coupling with $g > 0$. When is the product $\hbar/2$? If a linear phase $e^{ip_0 y/\hbar}$ is added, which mean system kick must be tracked separately?
2. **Finite noise–disturbance calculation (optional).** Use a CNOT with system controlling a meter initially $|0\rangle$, and system initially $|+y\rangle$. Put $A = \sigma_z$, $B = \sigma_x$, pointer $Z_M = \sigma_z$, and define $N = U^\dagger(\mathbb{1}\otimes Z_M)U - A\otimes\mathbb{1}$, $D = U^\dagger(B\otimes\mathbb{1})U - B\otimes\mathbb{1}$. Compute $\epsilon_A = \sqrt{\langle N^2\rangle}$, $\eta_B = \sqrt{\langle D^2\rangle}$, the input spreads and $|\langle[A, B]\rangle|/2$. Compare both $\epsilon_A\eta_B$ and $\epsilon_A\eta_B + \epsilon_A\Delta B + \Delta A\eta_B$ to that commutator quantity. This evaluates the finite Ozawa quantities, not a derivation of a universal inequality.
3. **Repeated selected tests (optional).** A qubit starts in $|0\rangle$ and evolves with $H = (\hbar\nu_R/2)\sigma_x$. After each of n equal intervals over time t, perform an ideal projective test and select outcome 0. Compute the probability of the full selected sequence and its limit as $n \to \infty$. Which experimental assumptions are essential?